Abbring & Van den Berg (2003), built up step by step

Plan

1. One duration: survival, hazard, integrated hazard
2. People differ: unobserved heterogeneity and dynamic selection
3. The mixed proportional hazard (MPH) model
4. Two durations racing: competing risks
5. Adding a treatment: potential outcomes indexed by treatment time, no anticipation
6. What we actually observe
7. The model of the paper (Model 1a)
8. Identification, step by step
9. Extensions: heterogeneous effects, multiple spells
10. Connections: DiD with staggered adoption, sequential exogeneity, traps

Running example (the paper's, Sections 1 and 2.4)

- Unemployed workers receive unemployment benefits.
- At a random time S during the spell, their **benefits are reduced** (from level w to $b < w$).
- Outcome: unemployment duration Y.
- Question: what is the causal effect of the benefits reduction on the **transition rate to work**?

1 One duration

Objects

- $Y \geq 0$: a continuous duration; y: a value.
- Density $f(y)$, CDF $F(y)$.
- Survival function $S(y) := \Pr(Y > y) = 1 - F(y)$: share still in the state at y.

1.1 Hazard rate

$$\begin{aligned}
\theta(y) &:= \lim_{dy \downarrow 0} \frac{\Pr(y \le Y < y + dy \mid Y \ge y)}{dy} && \text{[definition]} \\
&= \lim_{dy \downarrow 0} \frac{\Pr(y \le Y < y + dy)}{dy \ \Pr(Y \ge y)} && \text{[conditional probability: } \Pr(A \mid B) = \Pr(A \cap B)/\Pr(B)\text{]} \\
&= \lim_{dy \downarrow 0} \frac{F(y + dy) - F(y)}{dy} \cdot \frac{1}{S(y)} && \text{[}Y \text{ continuous: } \Pr(Y \ge y) = \Pr(Y > y)\text{]} \\
&= \frac{f(y)}{S(y)} && \text{[definition of the derivative: } F' = f\text{]} \qquad (1)
\end{aligned}$$

- Exit rate *among those still in the state* at y.
- $\theta(y)\, dy \approx$ probability of leaving in $[y, y + dy)$, given still there at y.
- A rate, not a probability: can exceed 1.

1.2 Integrated hazard and survival

$$\begin{aligned}
\theta(y) &= \frac{f(y)}{S(y)} && \text{[by (1)]} \\
&= -\frac{S'(y)}{S(y)} && \text{[}S = 1 - F\text{, so } S' = -f\text{]} \\
&= -\frac{d}{dy} \log S(y) && \text{[chain rule: } (\log S)' = S'/S\text{]}
\end{aligned}$$

$$\begin{aligned}
\Theta(y) := \int_0^y \theta(u)\, du &= -\big[\log S(u)\big]_0^y && \text{[fundamental theorem of calculus]} \\
&= -\log S(y) && \text{[}S(0) = 1\text{, so } \log S(0) = 0\text{]} \\
\implies \quad S(y) &= e^{-\Theta(y)} && \text{[exponentiate]} \qquad (2) \\
f(y) &= \theta(y)\, e^{-\Theta(y)} && \text{[}f = \theta S \text{ by (1)]} \qquad (3)
\end{aligned}$$

- Θ: integrated hazard.
- θ, Θ, S, f: four descriptions of the same distribution.

1.3 Example: constant hazard

$$\begin{aligned}
&\theta(y) = c \quad \implies \quad \Theta(y) = cy, \qquad S(y) = e^{-cy} && \text{[integrate; then (2)]} \\
&\Pr(Y > y + u \mid Y > y) = \frac{S(y + u)}{S(y)} && \text{[conditional probability; } \{Y > y + u\} \subset \{Y > y\}\text{]} \\
&\qquad\qquad = e^{-cu} && \text{[plug in } S\text{]}
\end{aligned}$$

- Does not depend on y: **memoryless**.
- θ decreasing in y: **negative duration dependence**. Increasing: **positive**.

1.4 Writing Y as a function of a standard random variable

Assume $\theta > 0$, so Θ is strictly increasing and invertible. Define $E := \Theta(Y)$. For $e \geq 0$:

$$\begin{aligned}
\Pr(E > e) &= \Pr\big(\Theta(Y) > e\big) && \text{[definition of } E\text{]} \\
&= \Pr\big(Y > \Theta^{-1}(e)\big) && \text{[apply } \Theta^{-1}\text{, strictly increasing]} \\
&= S\big(\Theta^{-1}(e)\big) && \text{[definition of } S\text{]} \\
&= e^{-\Theta(\Theta^{-1}(e))} = e^{-e} && \text{[by (2)]} \\
\implies \quad Y &= \Theta^{-1}(E), \qquad E \sim \mathrm{Exp}(1) && \text{[invert the definition of } E\text{]} \qquad (4)
\end{aligned}$$

- $E \sim \mathrm{Exp}(1)$ *whatever* θ is.
- Same idea as $F(Y) \sim U(0,1)$: indeed $E = -\log S(Y)$.
- Y = model part Θ applied to a standard random part E. Used in Sections 5 and 7.

2 Unobserved heterogeneity and dynamic selection

Model

- Individuals differ by an unobserved $V \geq 0$, $V \sim G$.
- Individual hazard: $\theta(y \mid V) = \lambda(y)\, V$. Baseline λ, integrated baseline $\Lambda(y) := \int_0^y \lambda$.
- So $\Theta(y \mid V) = \Lambda(y)V$ and, by (2), $\Pr(Y > y \mid V) = e^{-\Lambda(y)V}$.

2.1 Population survival and hazard

$$\begin{aligned}
\bar{S}(y) := \Pr(Y > y) &= \mathbb{E}\big[\Pr(Y > y \mid V)\big] && \text{[law of iterated expectations (LIE)]} \\
&= \mathbb{E}\big[e^{-\Lambda(y)V}\big] && \text{[by (2), conditional on } V\text{]} \\
\bar{\theta}(y) &= -\frac{\bar{S}'(y)}{\bar{S}(y)} && \text{[by (1) and } S' = -f\text{]} \\
&= \frac{\mathbb{E}\big[\lambda(y) V e^{-\Lambda(y)V}\big]}{\mathbb{E}\big[e^{-\Lambda(y)V}\big]} && \text{[differentiate under } \mathbb{E}\text{; } \Lambda' = \lambda\text{]} \\
&= \lambda(y)\, \frac{\mathbb{E}\big[V e^{-\Lambda(y)V}\big]}{\mathbb{E}\big[e^{-\Lambda(y)V}\big]} && \text{[linearity: } \lambda(y) \text{ is not random]} \qquad (5)
\end{aligned}$$

2.2 Who survives: the type distribution among survivors

$$\begin{aligned}
dG(v \mid Y > y) &= \frac{\Pr(Y > y \mid V = v)\, dG(v)}{\Pr(Y > y)} && \text{[Bayes' rule]} \\
&= \frac{e^{-\Lambda(y)v}\, dG(v)}{\mathbb{E}[e^{-\Lambda(y)V}]} && \text{[plug in the two survivals above]} \qquad (6) \\
\mathbb{E}[V \mid Y > y] &= \frac{\mathbb{E}[V e^{-\Lambda(y)V}]}{\mathbb{E}[e^{-\Lambda(y)V}]} && \text{[integrate } v \text{ against (6)]} \\
\implies \quad \bar{\theta}(y) &= \lambda(y)\, \mathbb{E}[V \mid Y > y] && \text{[compare with (5)]} \qquad (7)
\end{aligned}$$

- Survivors are the population **reweighted by their survival probability** $e^{-\Lambda(y)v}$.
- High-v people have lower weights: they leave first.

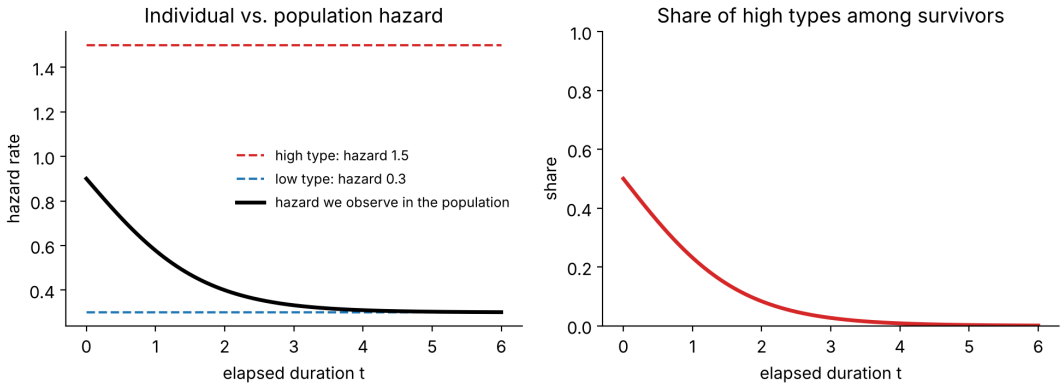
2.3 The mean type among survivors falls

Write $m(\ell) := \mathbb{E}[Ve^{-\ell V}]/\mathbb{E}[e^{-\ell V}]$, so $\mathbb{E}[V \mid Y > y] = m(\Lambda(y))$.

$$
\begin{aligned}
m'(\ell) &= \frac{-\mathbb{E}[V^2 e^{-\ell V}]\,\mathbb{E}[e^{-\ell V}] + \mathbb{E}[Ve^{-\ell V}]^2}{\mathbb{E}[e^{-\ell V}]^2} && \text{[quotient rule; differentiate under } \mathbb{E}\text{]} \\
&= -\Big(\mathbb{E}[V^2 \mid \cdot] - \mathbb{E}[V \mid \cdot]^2\Big) && \text{[each ratio is a survivor moment, as in (6)]} \\
&= -\operatorname{Var}(V \mid Y > y) \le 0 && \text{[definition of variance]}
\end{aligned}
$$

$$
\frac{d}{dy}\mathbb{E}[V \mid Y > y] = m'(\Lambda(y))\,\lambda(y) = -\lambda(y)\operatorname{Var}(V \mid Y > y) \quad \text{[chain rule]}
$$

- The population hazard falls *even if no individual hazard falls*. This is **dynamic selection**.
- Hence two explanations for a falling exit rate:
  - true duration dependence: $\lambda \downarrow$ (e.g. skill loss: a *trap*);
  - composition: $\mathbb{E}[V \mid Y > y] \downarrow$ (sorting on permanent heterogeneity).
- People still unemployed at month 6 are a selected group. Every comparison made at month 6 inherits this.



Two types with constant hazards 0.3 and 1.5 (equal shares). The population hazard falls from 0.9 to 0.3.

3 The mixed proportional hazard (MPH) model

Model

- $\theta(y \mid x, V) = \lambda(y)\,\varphi(x)\,V$, with $V \sim G$ and $V \perp X$.
- Three multiplicative pieces: duration (λ), observed (φ), unobserved (V).
- Normalize $\varphi(x^*) = 1$ and $\Lambda(y^*) = 1$ for some chosen x^*, y^*.
  - Reason: (λ, φ, V) and $(c\lambda, \varphi, V/c)$ give the same hazard.

3.1 Survival through the Laplace transform

Define the Laplace transform of G: $\mathcal{L}(a) := \mathbb{E}[e^{-aV}]$, $a \geq 0$.

$$
\begin{aligned}
S(y \mid x) &= \mathbb{E}\big[e^{-\varphi(x)\Lambda(y)V}\big] && \text{[LIE over } V\text{; (2); } V \perp X\text{]} \\
&= \mathcal{L}\big(\varphi(x)\Lambda(y)\big) && \text{[definition of } \mathcal{L}\text{]} \qquad (8) \\
\mathcal{L}'(a) &= -\mathbb{E}[Ve^{-aV}] < 0 && \text{[differentiate under } \mathbb{E}\text{]} \\
-\frac{\mathcal{L}'(a)}{\mathcal{L}(a)} &= \frac{\mathbb{E}[Ve^{-aV}]}{\mathbb{E}[e^{-aV}]} = \mathbb{E}[V \mid \text{survived to index } a] && \text{[as in (6)]} \\
\theta(y \mid x) &= \lambda(y)\varphi(x)\Big(-\frac{\mathcal{L}'}{\mathcal{L}}\Big)\big(\varphi(x)\Lambda(y)\big) && \text{[as in (5), now with } \varphi(x)\text{]} \qquad (9)
\end{aligned}
$$

- Data reveal one known function, $\mathcal{L}$, of a **single index** $\varphi(x)\Lambda(y)$.
- Facts used below:
  - $\mathcal{L}$ is strictly decreasing (if $\Pr(V > 0) > 0$);
  - $\mathcal{L}$ is real-analytic on $(0, \infty)$: known on an interval $\Rightarrow$ known everywhere;
  - $\mathcal{L}$ determines G uniquely.

3.2 Identification (Elbers & Ridder 1982) in three steps

Step 1: φ from short durations. Assume $\mathbb{E}V < \infty$.

$$
\begin{aligned}
\frac{\theta(y \mid x)}{\theta(y \mid x^*)} &= \varphi(x)\,\frac{(-\mathcal{L}'/\mathcal{L})(\varphi(x)\Lambda(y))}{(-\mathcal{L}'/\mathcal{L})(\Lambda(y))} && \text{[by (9); } \lambda(y) \text{ cancels; } \varphi(x^*) = 1\text{]} \\
&\xrightarrow{y \downarrow 0} \varphi(x)\,\frac{-\mathcal{L}'(0)/\mathcal{L}(0)}{-\mathcal{L}'(0)/\mathcal{L}(0)} = \varphi(x) && \text{[}\Lambda(y) \to 0\text{; continuity; } -\mathcal{L}'(0) = \mathbb{E}V < \infty\text{]}
\end{aligned}
$$

- At $y \approx 0$ nobody has been selected out yet, so the ratio is free of selection.
- Uses short durations, where data are thin.

Step 2: G at the normalization date.

$$
S(y^* \mid x) = \mathcal{L}\big(\varphi(x)\big) \qquad \text{[(8) with } \Lambda(y^*) = 1\text{]}
$$

- $\varphi(x)$ known from step 1. Varying x traces $\mathcal{L}$ on an interval.
- Analytic $\Rightarrow$ $\mathcal{L}$ known on $(0, \infty) \Rightarrow$ G known.

Step 3: baseline.

$$
\Lambda(y) = \frac{\mathcal{L}^{-1}\big(S(y \mid x)\big)}{\varphi(x)} \qquad \text{[invert (8); } \mathcal{L} \text{ strictly decreasing]}
$$

What was assumed: proportionality, $V \perp X$, $\mathbb{E}V < \infty$, variation in $\varphi(x)$.

4 Two durations racing: competing risks

Set-up

- Two latent durations: Y (exit to work) and S (benefits reduction).
- Only the **first** event is seen, plus which one it was.

- MPH for each, with **correlated** unobservables $V = (V_Y, V_S) \sim G$, $V \perp X$:

$$\theta_Y(t \mid x, V) = \lambda_Y(t)\varphi_Y(x)V_Y, \qquad \theta_S(t \mid x, V) = \lambda_S(t)\varphi_S(x)V_S.$$

- Given (x, V): $Y \perp S$. All dependence runs through V.
- Bivariate Laplace transform: $\mathcal{L}(a, b) := \mathbb{E}[e^{-aV_Y - bV_S}]$; partials $\mathcal{L}_a, \mathcal{L}_b, \mathcal{L}_{ab}$.
- Indices: $a_t(x) := \varphi_Y(x)\Lambda_Y(t)$, $b_t(x) := \varphi_S(x)\Lambda_S(t)$.

4.1 Joint survival

$$\begin{aligned}
\Pr(Y > y, S > s \mid x) &= \mathbb{E}\big[\Pr(Y > y, S > s \mid x, V)\big] && \text{[LIE over } V\text{; } V \perp X\text{]} \\
&= \mathbb{E}\big[\Pr(Y > y \mid x, V)\Pr(S > s \mid x, V)\big] && \text{[}Y \perp S \mid (x, V)\text{]} \\
&= \mathbb{E}\big[e^{-\varphi_Y(x)\Lambda_Y(y)V_Y}\, e^{-\varphi_S(x)\Lambda_S(s)V_S}\big] && \text{[(2) for each]} \\
&= \mathcal{L}\big(\varphi_Y(x)\Lambda_Y(y),\, \varphi_S(x)\Lambda_S(s)\big) && \text{[definition of } \mathcal{L}\text{]} \qquad (10)
\end{aligned}$$

4.2 What the data reveal

Probability that nothing has happened by t (the diagonal $y = s = t$):

$$P_0(t \mid x) := \Pr(Y > t, S > t \mid x) = \mathcal{L}\big(a_t(x), b_t(x)\big) \qquad \text{[(10) at } y = s = t\text{]} \qquad (11)$$

Cause-specific hazard of exit (rate of Y happening first, among those with nothing yet):

$$\begin{aligned}
h_Y(t \mid x) &:= \lim_{dt \downarrow 0} \frac{\Pr(t \le Y < t + dt,\ S > t \mid x)}{dt\ P_0(t \mid x)} && \text{[definition]} \\
&= \frac{-\partial_y \Pr(Y > y, S > t \mid x)\big|_{y=t}}{P_0(t \mid x)} && \text{[derivative of a survival in } y\text{]} \\
&= \frac{-\mathcal{L}_a(a_t, b_t)\,\varphi_Y(x)\lambda_Y(t)}{\mathcal{L}(a_t, b_t)} && \text{[chain rule on (10); } \Lambda_Y' = \lambda_Y\text{]} \\
&= \lambda_Y(t)\varphi_Y(x)\,\mathbb{E}[V_Y \mid R_t, x] && \text{[}-\mathcal{L}_a/\mathcal{L} = \text{survivor mean, as in (6)]} \qquad (12)
\end{aligned}$$

where $R_t := \{Y > t, S > t\}$ is the **risk set**. Symmetrically, $h_S(t \mid x) = \lambda_S(t)\varphi_S(x)\mathbb{E}[V_S \mid R_t, x]$.

4.3 Where the dependence of V_Y, V_S lives

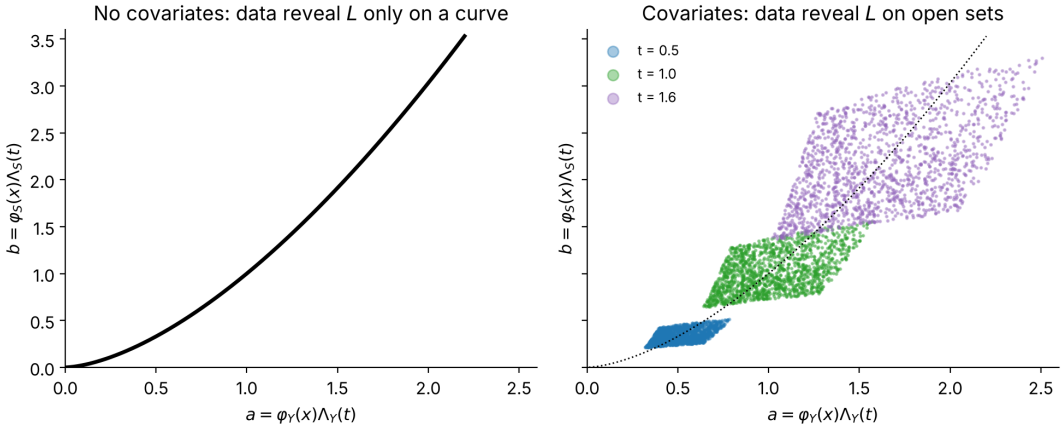
Write $w(v) \propto e^{-av_Y - bv_S}$ (survivor weights) and $\mathbb{E}_w$ for expectations under them.

$$\begin{aligned}
\partial_a \log \mathcal{L} &= \frac{\mathcal{L}_a}{\mathcal{L}} = -\mathbb{E}_w[V_Y] && \text{[chain rule; differentiate under } \mathbb{E}\text{]} \\
\partial_b \partial_a \log \mathcal{L} &= \frac{\mathcal{L}_{ab}}{\mathcal{L}} - \frac{\mathcal{L}_a \mathcal{L}_b}{\mathcal{L}^2} && \text{[quotient rule]} \\
&= \mathbb{E}_w[V_Y V_S] - \mathbb{E}_w[V_Y]\mathbb{E}_w[V_S] && \text{[}\mathcal{L}_{ab} = \mathbb{E}[V_Y V_S e^{\cdots}]\text{, } \mathcal{L}_b/\mathcal{L} = -\mathbb{E}_w[V_S]\text{]} \\
&= \mathrm{Cov}(V_Y, V_S \mid R_t, x) && \text{[definition of covariance; at } (a, b) = (a_t, b_t)\text{]} \qquad (13)
\end{aligned}$$

- $V_Y \perp V_S \iff \mathcal{L}(a, b) = \mathcal{L}_Y(a)\mathcal{L}_S(b) \iff \log P_0$ additively separable in the two indices.
- Dependence (selection) = an **interaction** of the two indices in $\log P_0$.

4.4 The problem: Tsiatis (1975)

- Without structure, any competing-risks data can be generated by *independent* Y, S.
- Geometry, without covariates:
  - the data reveal $\mathcal{L}$ only at points $(\Lambda_Y(t), \Lambda_S(t))$, i.e. along a **curve**;
  - a curve cannot pin down a function of two variables, so the dependence is not identified.
- With covariates moving φ_Y and φ_S separately: points $(a_t(x), b_t(x))$ fill **open sets**.



4.5 Identification (Prop. 2) in three steps

Assumptions. (A3) $\{(\varphi_Y(x), \varphi_S(x))\}$ contains an open set of $\mathbb{R}^2$. (A4) $\mathbb{E}V_Y, \mathbb{E}V_S < \infty$. Normalize $\varphi_Y(x^*) = \varphi_S(x^*) = 1$, $\Lambda_Y(t^*) = \Lambda_S(t^*) = 1$.

Step 1: φ_Y, φ_S**.**

$$\frac{h_Y(t \mid x)}{h_Y(t \mid x^*)} = \varphi_Y(x)\,\frac{\mathbb{E}[V_Y \mid R_t, x]}{\mathbb{E}[V_Y \mid R_t, x^*]} \qquad \text{[by (12); } \lambda_Y(t) \text{ cancels]}$$

$$\xrightarrow{t\downarrow 0} \varphi_Y(x)\,\frac{\mathbb{E}V_Y}{\mathbb{E}V_Y} = \varphi_Y(x) \qquad \text{[risk set} \to \text{population; } V \perp X\text{; (A4)]}$$

Same for φ_S with h_S. Uses *which* event happened.

Step 2: G**, including the dependence.**

$$P_0(t^* \mid x) = \mathcal{L}\big(\varphi_Y(x), \varphi_S(x)\big) \qquad \text{[(11) with } \Lambda_Y(t^*) = \Lambda_S(t^*) = 1]$$

- Known points fill an open set (A3) $\Rightarrow$ $\mathcal{L}$ known there.
- Analytic $\Rightarrow$ known on $(0, \infty)^2$ $\Rightarrow$ joint G known.

Step 3: baselines. For each t, $(\Lambda_Y(t), \Lambda_S(t))$ is the unique (a, b) with

$$P_0(t \mid x) = \mathcal{L}\big(\varphi_Y(x)a,\ \varphi_S(x)b\big) \quad \text{for all } x. \qquad \text{[(11); } \mathcal{L}, \varphi\text{'s known]}$$

- Uniqueness: two solutions would give $\mathcal{L}(u, v) = \mathcal{L}(ku, mv)$ on $(0, \infty)^2$, $(k, m) \neq (1, 1)$.
- Iterating, $\mathcal{L}$ would be constant, contradicting strict monotonicity.
- These three steps are our reconstruction; the paper cites Abbring & Van den Berg (2003, JRSSB).

5 Adding a treatment

5.1 Potential outcomes indexed by treatment time

- Treatment = benefits reduction at date $s \in [0, \infty]$; $s = \infty$: never.
- One potential outcome per date: $Y^*(s)$, with hazard $\theta_{Y^*(s)}$, integrated hazard $\Theta_{Y^*(s)}$.
- By (4): $Y^*(s) = \Theta^{-1}_{Y^*(s)}(E)$, $E \sim \text{Exp}(1)$.
- Actual treatment date S; observed outcome $Y = Y^*(S)$.

5.2 No anticipation (Assumption 1)

$$\Theta_{Y^*(s)}(y) = \Theta_{Y^*(s')}(y) \qquad \text{for all } y \leq \min\{s, s'\}.$$

Take $s' = \infty$ and differentiate in y:

$$\theta_{Y^*(s)}(y) = \theta_{Y^*(\infty)}(y) \qquad \text{for } y < s. \qquad \text{[derivative of both sides]}$$

- Before treatment, everyone has the **untreated** hazard, whatever their future date.
- Economic content: people do not act on knowledge of their own future reduction date.
- Violated if, e.g., the date is announced in advance and people search harder before it.
- Not ruled out: reacting to the *policy* (e.g. the rate at which reductions happen).
  - This reaction is in every $Y^*(s)$.
  - So the effect is "treated at s vs. later, under this policy", not "policy vs. no policy".
- Prop. 1: without more structure, data cannot distinguish (i) causal effects from selection, nor (ii) anticipation from no anticipation. Hence Assumption 1 is maintained, and selection must be modelled.

6 What we actually observe

The censoring

- S is seen only if $S < Y$: if the worker finds a job first, the reduction date is never seen.
- Paper's data (per x): $Q_S(y, s) := \Pr(Y > y, S > s, Y > S)$ and $Q_Y(y) := \Pr(Y > y, Y < S)$. Assume $\Pr(Y = S) = 0$.

Pre-treatment objects

$$\begin{aligned} P_0(t) &= \Pr(Y > t, S > t, Y > S) + \Pr(Y > t, S > t, Y < S) && \text{[partition by } Y \gtrless S\text{; } \Pr(Y = S) = 0\text{]} \\ &= Q_S(t, t) + Q_Y(t) && \text{[} Y > t, Y < S \text{ implies } S > t\text{]} \\ h_Y(t) &= \frac{-Q'_Y(t)}{P_0(t)} && \text{[} -Q'_Y\text{: density of untreated exits]} \\ h_S(t) &= \frac{-\partial_s Q_S(y, s)\big|_{y \downarrow t,\, s=t}}{P_0(t)} && \text{[density of } S \text{ at } t\text{, still unemployed]} \end{aligned}$$

Post-treatment object: survival of the cohort treated at s. For $y > s$:

$$\begin{aligned}
-\partial_s Q_S(y,s) &= \lim_{ds\downarrow 0} \frac{\Pr(Y > y,\ s \le S < s+ds,\ Y > S)}{ds} && \text{[derivative of } Q_S \text{ in } s\text{]} \\
&= \lim_{ds\downarrow 0} \frac{\Pr(Y > y,\ s \le S < s+ds)}{ds} && \text{[}Y > y > S \text{ implies } Y > S\text{]} \\
P_1(y \mid s) &:= \Pr(Y > y \mid S = s,\, Y > s) && \text{[definition]} \\
&= \frac{\partial_s Q_S(y,s)}{\lim_{y'\downarrow s} \partial_s Q_S(y',s)} && \text{[cond. probability; } ds \text{ cancels]} \qquad (14)
\end{aligned}$$

- (P_0, h_Y, h_S) = competing-risks data, as in Section 4.
- P_1 = the only place where the treatment effect can show up.

7 The model of the paper (Model 1a)

Assumption 2 (selection on observables and unobservables): $Y^*(\cdot) \perp S \mid (X, V)$, with V time-invariant.

Model 1a

$$\theta_Y(t \mid s, x, V) = \begin{cases} \lambda_Y(t)\varphi_Y(x)V_Y & t \le s \\ \lambda_Y(t)\varphi_Y(x)V_Y\,\delta(t \mid s, x) & t > s \end{cases}$$
$$\theta_S(t \mid x, V) = \lambda_S(t)\varphi_S(x)V_S, \qquad (V_Y, V_S) \sim G, \quad V \perp X.$$

- δ: **the treatment effect**, a multiplier on the exit rate after treatment.
  - May depend on t, s (hence on time since treatment $t - s$), and x; not on V.
- No anticipation is built in: δ only acts for $t > s$.
- Selection on unobservables: V_Y and V_S may be dependent in any way.
- The same x in both hazards: **no exclusion restriction**.
- $\lambda_Y, \lambda_S, \varphi_Y, \varphi_S, G, \delta$ all nonparametric.
- Plus (A3), (A4) from Section 4.

Integrated outcome hazard, with $D(y \mid s, x) := \int_s^y \lambda_Y(\tau)\delta(\tau \mid s, x)\, d\tau$:

$$\begin{aligned}
\Theta_Y(y \mid s, x, V) &= \int_0^s \lambda_Y\varphi_Y V_Y\, d\tau + \int_s^y \lambda_Y\varphi_Y V_Y\delta\, d\tau && \text{[split the integral at } s\text{]} \\
&= \varphi_Y(x)V_Y\big[\Lambda_Y(s) + D(y \mid s, x)\big] \quad (y > s) && \text{[linearity of the integral]} \qquad (15)
\end{aligned}$$

Treatment timing is random given (x, V)

$$\begin{aligned}
E_S := \Theta_S(S \mid x, V) = \varphi_S(x)V_S\Lambda_S(S) \ &\sim\ \text{Exp}(1), \quad E_S \perp (x, V) && \text{[by (4), conditional on } (x, V)\text{]} \\
&\Longrightarrow \quad S = \Lambda_S^{-1}\Big(\frac{E_S}{\varphi_S(x)V_S}\Big) && \text{[invert]}
\end{aligned}$$

- E_S moves S around without entering Y's hazard (Assumption 2).
- Like an instrument, except that what is excluded is an unobserved **shock**, not an observed variable.

8 Identification, step by step

8.1 Step A: the pre-treatment data are the competing-risks model of Section 4

$$\begin{aligned} P_0(t \mid x) &= \mathbb{E}\big[\Pr(Y > t, S > t \mid x, V)\big] && \text{[LIE; } V \perp X\text{]} \\ &= \mathbb{E}\big[e^{-\varphi_Y(x)\Lambda_Y(t)V_Y}\, e^{-\varphi_S(x)\Lambda_S(t)V_S}\big] && \text{[Assumption 2; before } S\text{, } \delta \text{ absent (no anticipation)]} \\ &= \mathcal{L}\big(a_t(x), b_t(x)\big) && \text{[as in (11)]} \end{aligned}$$

- Same for h_Y, h_S. So Prop. 2 (Section 4.5) applies: $\varphi_Y, \varphi_S, \Lambda_Y, \Lambda_S, G$ are identified.
- This uses no post-treatment data.

8.2 Step B: who gets treated at s

Risk set at s:

$$\begin{aligned} dG_{R_s}(v) &= \frac{\Pr(R_s \mid x, v)\, dG(v)}{\Pr(R_s \mid x)} && \text{[Bayes' rule]} \\ &\propto e^{-a_s v_Y - b_s v_S}\, dG(v) && \text{[as in Step A]} \end{aligned}$$

Cohort treated at s (in the risk set and treated in $[s, s+ds)$):

$$\begin{aligned} dG_s(v) &\propto \Pr(s \le S < s + ds \mid R_s, x, v)\, dG_{R_s}(v) && \text{[Bayes' rule]} \\ &\propto \lambda_S(s)\varphi_S(x) v_S\, ds\, dG_{R_s}(v) && \text{[definition of the hazard } \theta_S\text{]} \\ &\propto v_S\, e^{-a_s v_Y - b_s v_S}\, dG(v) && \text{[drop factors not depending on } v\text{]} \end{aligned} \tag{16}$$

- Treated cohort = risk set **reweighted by the treatment intensity** v_S.
- Fully known after Step A.

8.3 Step C: the selection problem at the moment of treatment

Exit hazard of the just-treated vs. the not-yet-treated, at s:

$$\begin{aligned} \text{just treated:} \quad & \lambda_Y(s)\varphi_Y(x)\delta(s \mid s, x)\, \mathbb{E}_{G_s}[V_Y] && \text{[Model 1a, averaged over (16)]} \\ \text{not yet treated:} \quad & \lambda_Y(s)\varphi_Y(x)\, \mathbb{E}_{R_s}[V_Y] && \text{[(12)]} \end{aligned}$$

$$\begin{aligned} \mathbb{E}_{G_s}[V_Y] &= \frac{\mathbb{E}_{R_s}[V_Y V_S]}{\mathbb{E}_{R_s}[V_S]} && \text{[(16): } G_s = G_{R_s} \text{ reweighted by } v_S\text{]} \\ &= \mathbb{E}_{R_s}[V_Y] + \frac{\mathrm{Cov}_{R_s}(V_Y, V_S)}{\mathbb{E}_{R_s}[V_S]} && \text{[}\mathbb{E}[XY] = \mathbb{E}X\mathbb{E}Y + \mathrm{Cov}(X,Y)\text{]} \\ \frac{\text{just treated}}{\text{not yet treated}} &= \delta(s \mid s, x)\Big(1 + \frac{\mathrm{Cov}_{R_s}(V_Y, V_S)}{\mathbb{E}_{R_s}[V_Y]\mathbb{E}_{R_s}[V_S]}\Big) && \text{[divide; } \lambda_Y\varphi_Y \text{ cancel]} \\ &= \delta(s \mid s, x) \cdot \frac{\mathcal{L}\,\mathcal{L}_{ab}}{\mathcal{L}_a\,\mathcal{L}_b}\Big|_{(a_s, b_s)} && \text{[survivor moments from Section 4.3]} \end{aligned}$$

- Observed ratio = causal effect × **selection factor**.
- Selection factor = 1 iff V_Y, V_S uncorrelated in the risk set: then timing is as good as random at s.
- The selection factor is known from Step A, so $\delta(s \mid s, x)$ is identified.

8.4 Step D: beyond the moment of treatment (Prop. 3)

For $y > s$, let $g(y, s \mid x)$ be the density of "treated at s and still unemployed at y", and $D := D(y \mid s, x)$:

$$\begin{aligned} g(y, s \mid x) &= \mathbb{E}\big[f_S(s \mid x, V)\ \Pr(Y > y \mid S = s, x, V)\big] && \text{[LIE; Assumption 2]} \\ &= \mathbb{E}\big[\lambda_S(s)\varphi_S V_S e^{-b_s V_S} \cdot e^{-\varphi_Y V_Y [\Lambda_Y(s)+D]}\big] && \text{[(3); (2) with (15)]} \\ &= \lambda_S(s)\varphi_S\, \mathbb{E}\big[V_S\, e^{-(a_s+\varphi_Y D)V_Y - b_s V_S}\big] && \text{[linearity; } a_s = \varphi_Y \Lambda_Y(s)\text{]} \\ &= -\lambda_S(s)\varphi_S(x)\, \mathcal{L}_b\big(a_s + \varphi_Y(x)D,\ b_s\big) && \text{[}\mathcal{L}_b = -\mathbb{E}[V_S e^{-aV_Y - bV_S}]\text{]} \end{aligned}$$

Divide by the same expression at $y \downarrow s$ (where $D = 0$), as in (14):

$$\begin{aligned} P_1(y \mid s, x) &= \frac{\mathcal{L}_b\big(a_s + \varphi_Y(x)D(y \mid s, x),\ b_s\big)}{\mathcal{L}_b(a_s, b_s)} && \text{[the } \lambda_S\varphi_S \text{ cancel]} \\ &= \mathbb{E}_{G_s}\big[e^{-\varphi_Y(x)V_Y D(y|s,x)}\big] && \text{[rewrite with weights (16)]} \end{aligned} \tag{17}$$

- Left side: data. Right side: everything known from Step A, except D.
- Strictly decreasing in $D \Rightarrow$ invert for $D(y \mid s, x)$.
- Then $\delta(y \mid s, x) = \partial_y D(y \mid s, x)/\lambda_Y(y)$ [fundamental theorem of calculus].
- This is the paper's eq. (10), divided by its value at $y \downarrow s$.

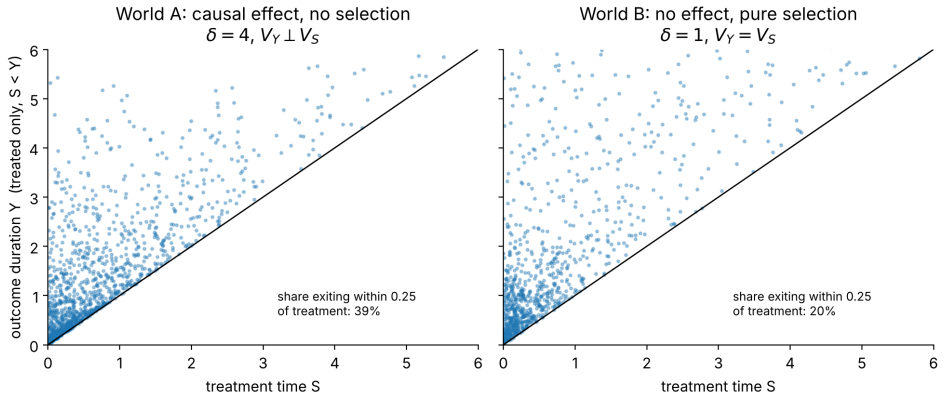
Two selection problems, both handled by knowing G

1. Selection *into* treatment at s: the v_S-tilt in (16).
2. Selection *out* after s: in (17), survivors of the treated cohort drift to low V_Y, as in Section 2.
   - Present whenever V_Y varies, even with random timing.
   - So naive hazard ratios at later y are not δ.

8.5 Summary: what is used where

Assumption	Where it is used
No anticipation	Step A: before S there is no δ, so a clean competing-risks problem
Assumption 2	Steps A, B, D: independence given V; treated = risk set reweighted by v_S
MPH, $V \perp X$	Step A: everything is $\mathcal{L}$ of indices
$\mathbb{E}V < \infty$ (A4)	Step A, step 1: short-duration hazard ratios
Open set (A3)	Step A, step 2: trace $\mathcal{L}$ in 2-D $\Rightarrow$ dependence of V_Y, V_S

- Not used: an excluded instrument, selection on observables, parametric forms, multiple spells.



Intuition. Left: causal effect, no selection; exits bunch right after treatment at every S (local). Right: no effect, pure selection; S and Y correlated overall (global).